

SBERT Classifier Comparison

Sections 3.4 and 3.5 | Research experiment results

Encoder: sentence-transformers/all-mpnet-base-v2. Frozen, normalized 768-dimensional embeddings of existing English translations. PCA reduces the text representation to 64 dimensions.

Dataset: 2,252 synthetic cardiac records. Stratified split: 1,801 training and 451 test rows; seed 42. Every comparison uses the same test rows. PCA and patient-feature preprocessing are fitted on training rows only.

Text-only comparison

Text size	Classifier	Accuracy	Precision*	F1*
768-D	Logistic Regression	58.09%	53.49%	54.85%
768-D	Hist Gradient Boosting	57.43%	53.29%	47.98%
768-D	Random Forest	55.65%	55.53%	51.08%
PCA-64	Logistic Regression	55.65%	50.68%	51.78%
PCA-64	Hist Gradient Boosting	56.98%	51.39%	47.34%
PCA-64	Random Forest	54.32%	47.86%	46.53%

Text plus patient features

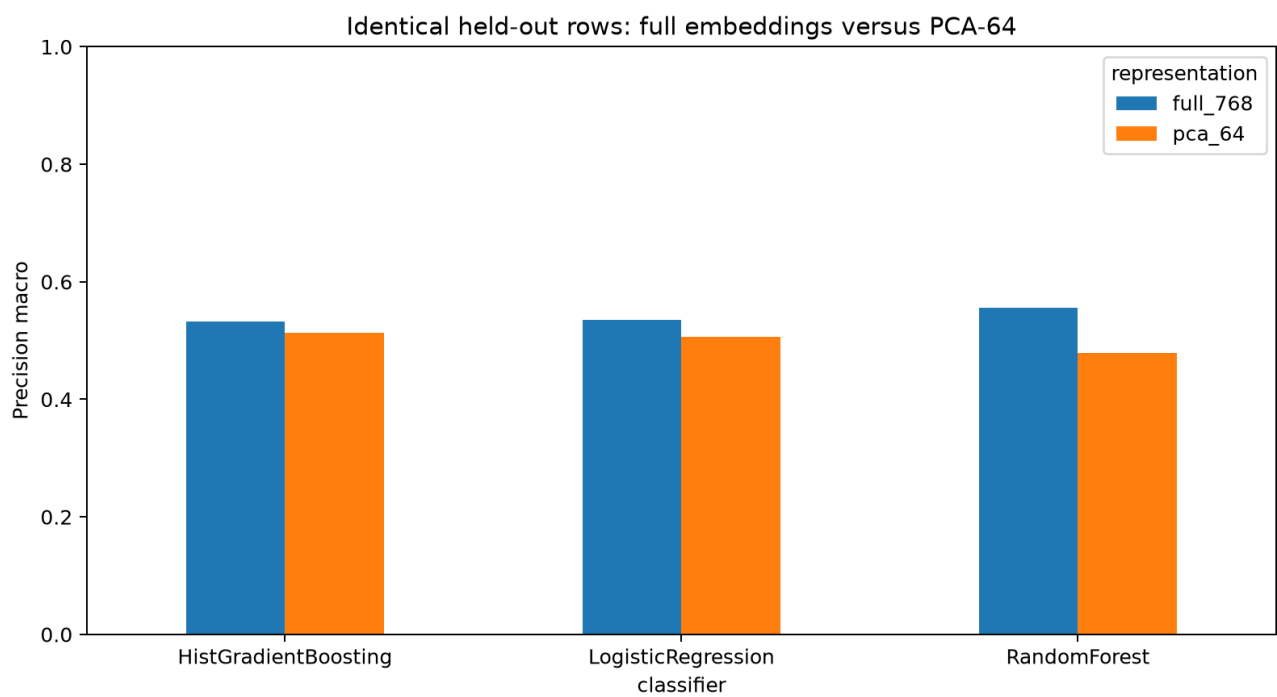
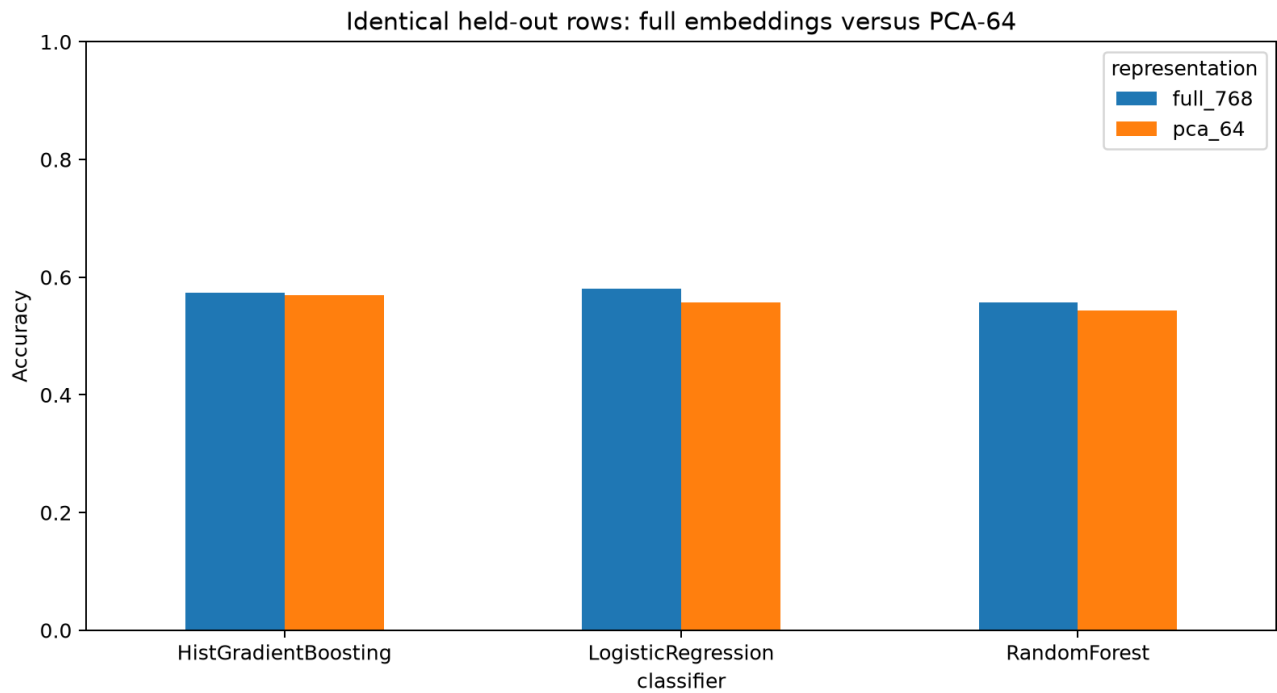
Text size	Classifier	Accuracy	Precision*	F1*
768-D	Logistic Regression	82.48%	77.03%	78.68%
768-D	Hist Gradient Boosting	82.93%	82.08%	76.97%
768-D	Random Forest	72.73%	70.12%	64.37%
PCA-64	Logistic Regression	82.26%	76.58%	78.28%
PCA-64	Hist Gradient Boosting	80.71%	81.54%	77.27%
PCA-64	Random Forest	80.71%	79.13%	73.06%

*Precision and F1 are macro-averaged. Fusion adds patient features to the stated text dimension. PCA retains 89.71% of training embedding variance.

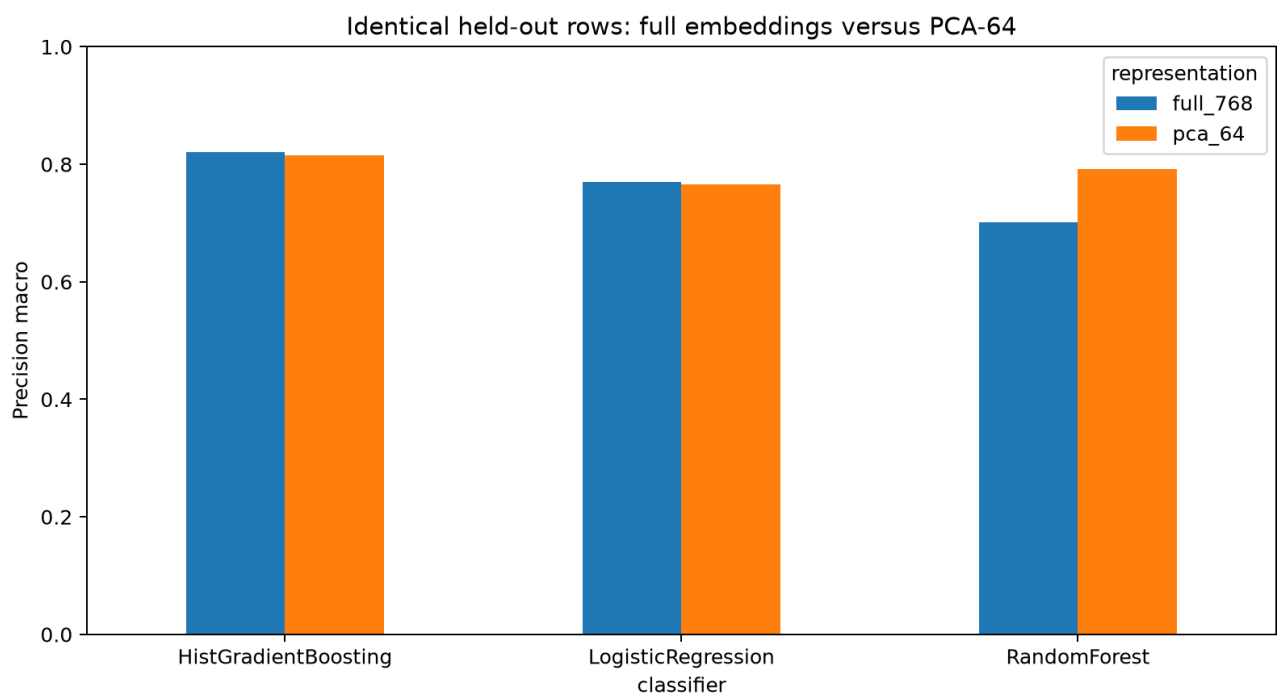
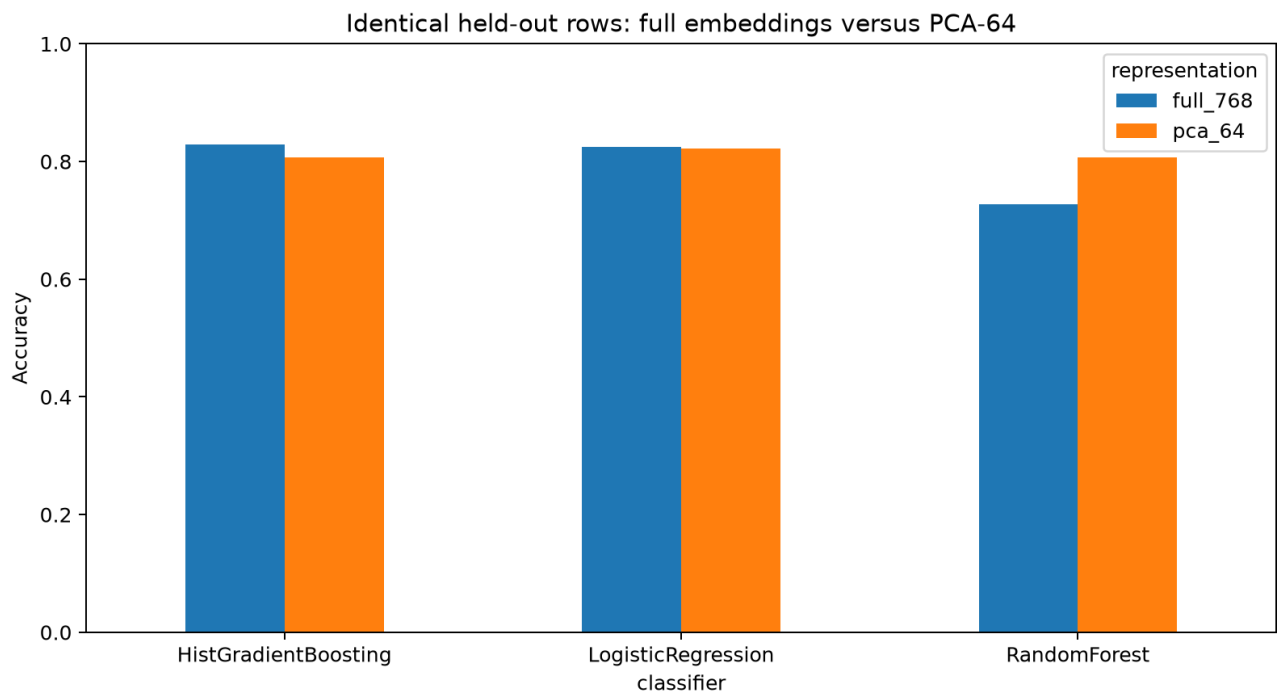
Interpretation and Section 3.6

These are single-split synthetic-data results, not clinical validation or an evaluation of live translation. Repeated complaint phrases may cross the random row split. Section 3.6 remains pending: the article has only a heading and no published comparator list. No literature scores have been invented.

Text-only performance

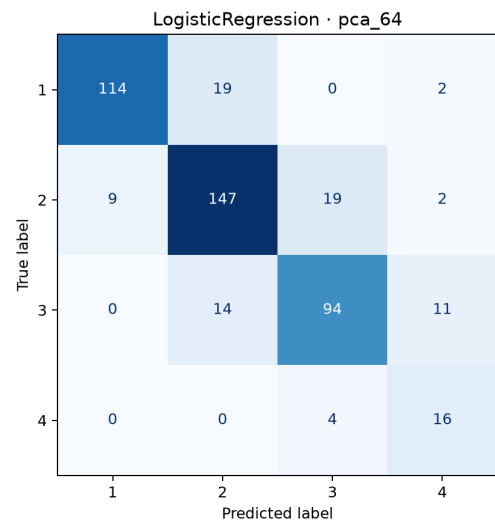
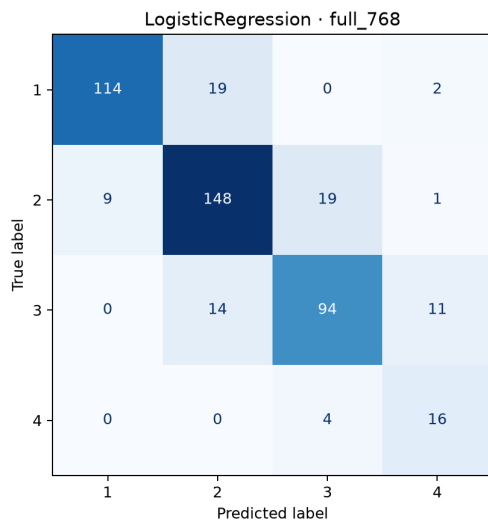
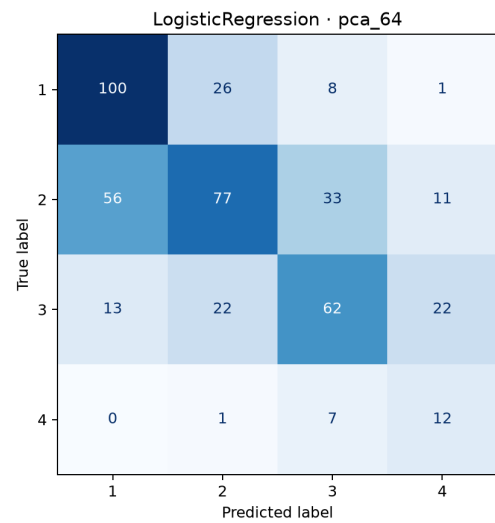
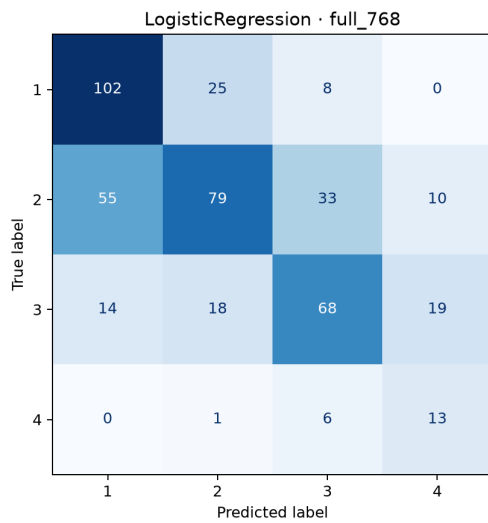


Performance with patient features



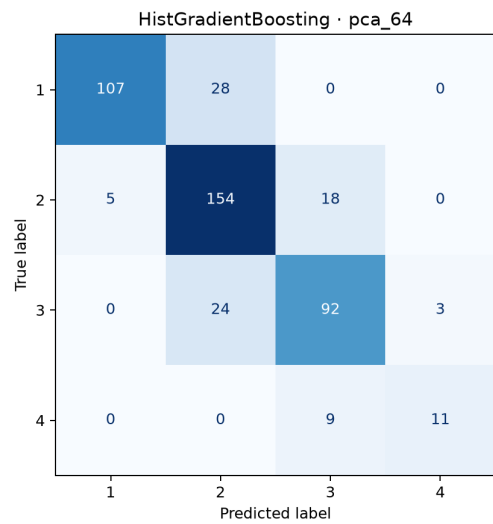
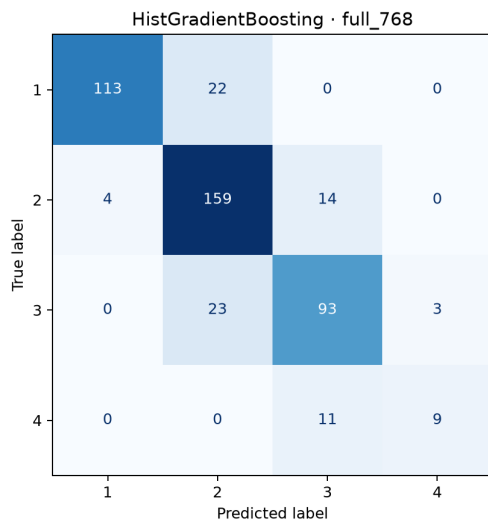
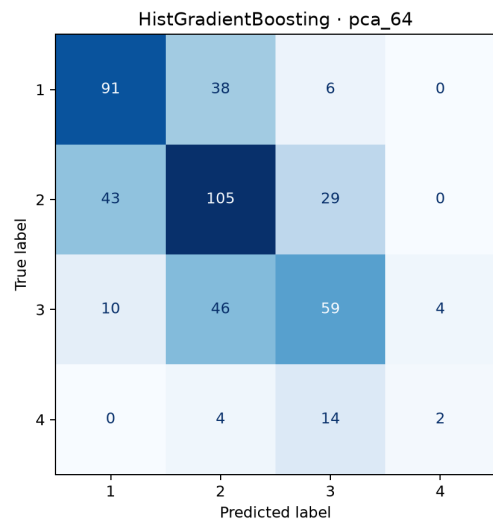
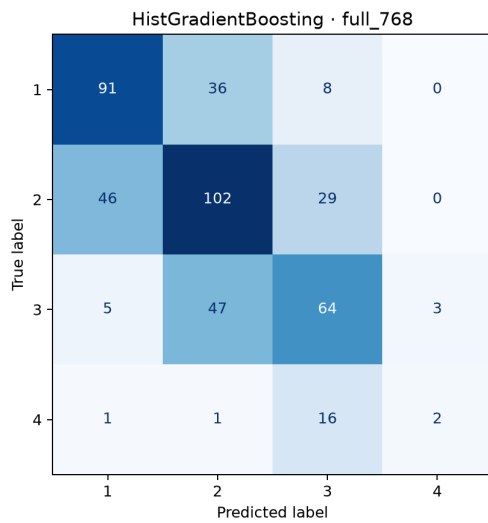
Logistic Regression: confusion matrices

Rows show true triage levels; columns show predictions. Each matrix contains the same 451 test records. Top row: text only. Bottom row: text plus patient features. Left: original 768-D; right: PCA-64.



Hist Gradient Boosting: confusion matrices

Rows show true triage levels; columns show predictions. Each matrix contains the same 451 test records. Top row: text only. Bottom row: text plus patient features. Left: original 768-D; right: PCA-64.



Random Forest: confusion matrices

Rows show true triage levels; columns show predictions. Each matrix contains the same 451 test records. Top row: text only. Bottom row: text plus patient features. Left: original 768-D; right: PCA-64.

